MCU

File: mcu.kicad_sch

Power

File: power.kicad_sch

Outputs

File: outputs.kicad_sch

Inputs

File: inputs.kicad_sch

Connectors and CAN

File: connectors.kicad_sch

Wideband controller

File: wideband.kicad_sch

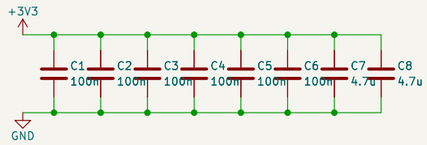
Miata ECU daughterboard rev 0.5 (on-board wideband controller, flashed through the main MCU; 5 V low-side drive, fused logic supply). Block sheets connect through global labels; GND, +3V3, +5V and +12V are power sym

Sheet: /
File: miata_ecu_daughter.kicad_sch

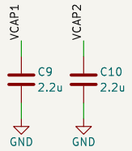
Title: Miata ECU daughterboard

Size: A4	Date: 2026-10-30	**Rev: 0.5**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/7

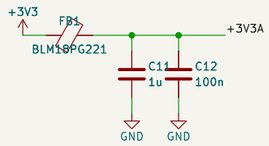
Decoupling: one 100 nF per VDD pin, 2 x 4.7 uF bulk



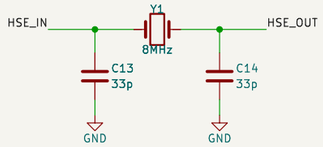
Core regulator caps



Analog supply filter

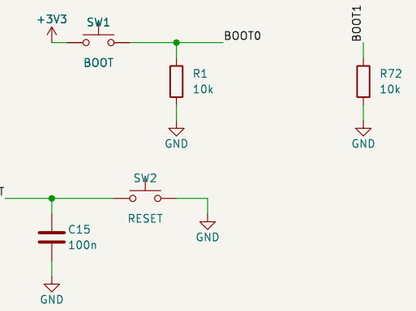


8 MHz HSE crystal

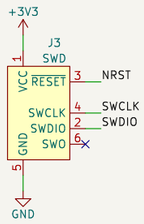


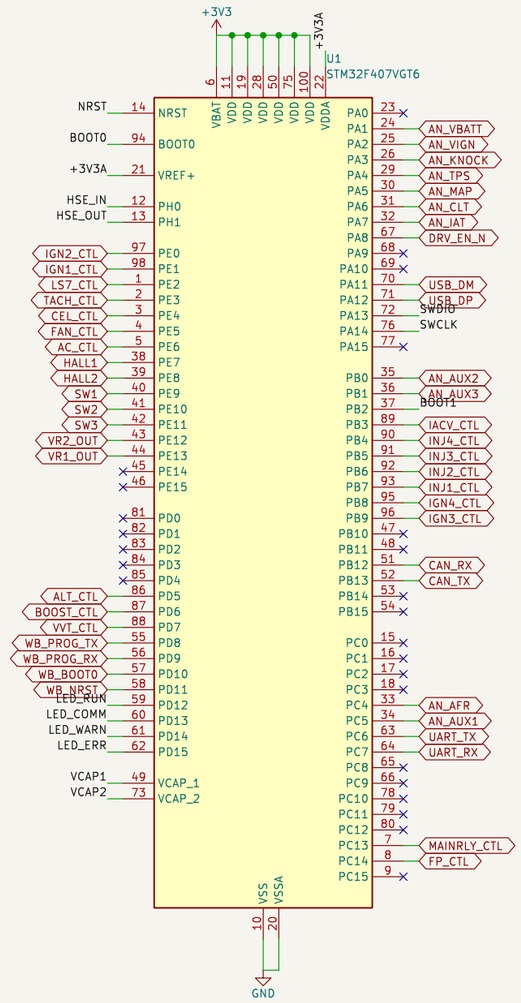
BOOT0 (hold for DFU) and reset

BOOT1 (PB2) low: DFU needs BOOT0 = 1, BOOT1 = 0

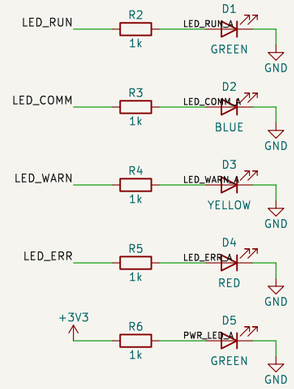


SWD (Tag-Connect TC2030-NL pads)





Status LEDs



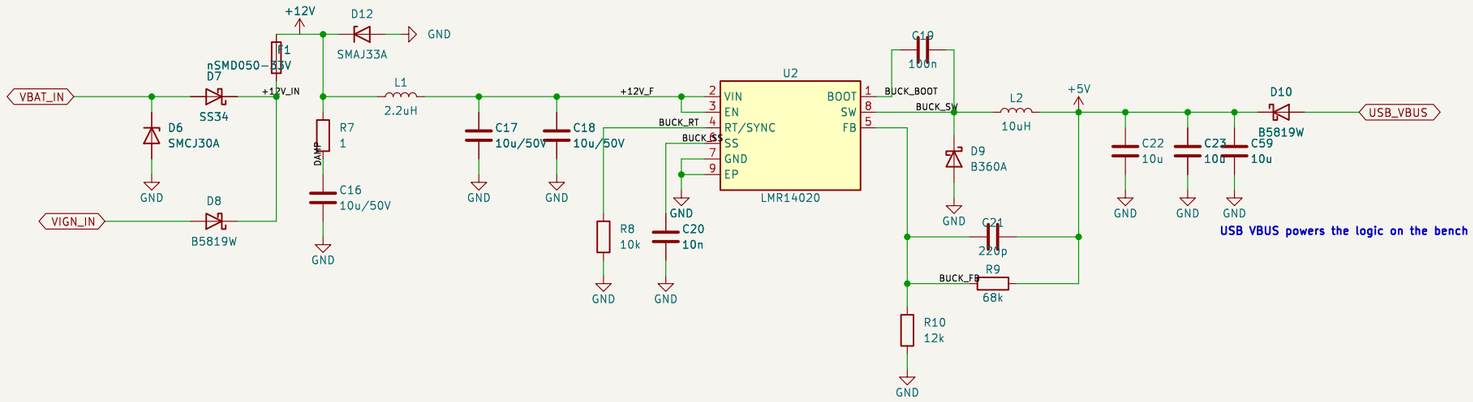
Generated from scripts/design.py; netlist identical to the routed board

Sheet: /MCU/
File: mcu.kicad_sch

Title: Miata ECU daughterboard — MCU

Size: A3	Date: 2026-10-31	**Rev: 0.5b**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/7

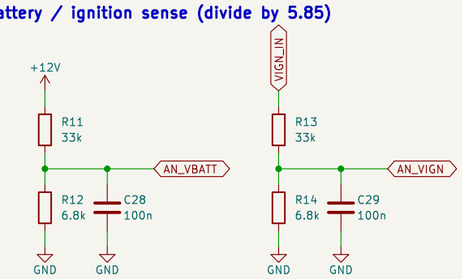
Input protection (fused, TVS-clamped logic supply) and 5 V buck (LMR14020, 5 V / 2 A)



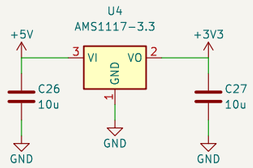
USB VBUS powers the logic on the bench

Fsw ~ 1 MHz (RT 10k); Vout = 0.75 V x (1 + 68k/12k) = 5.0 V

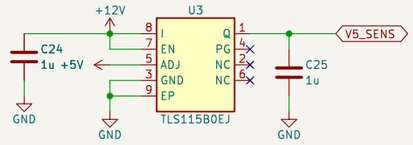
Battery / ignition sense (divide by 5.85)



3.3 V LDO



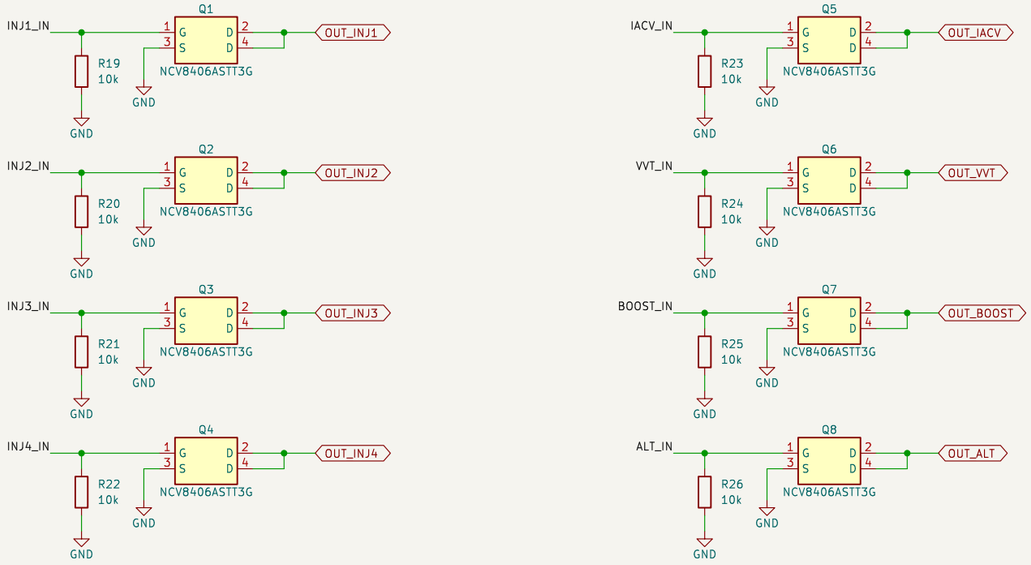
Protected 5 V sensor supply (TLS115 tracks +5V)



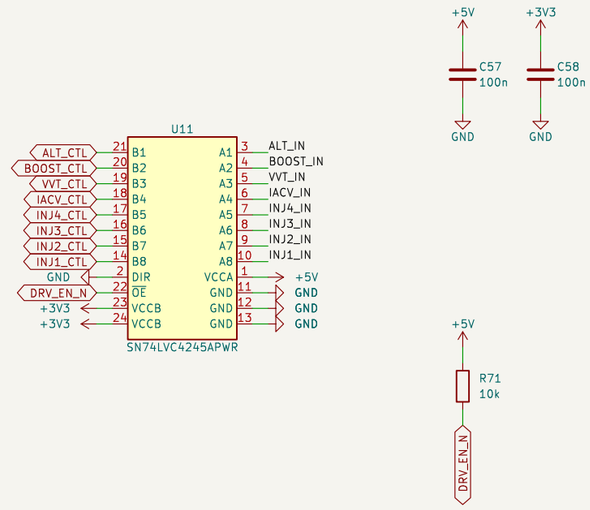
Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Power/ File: power.kicad_sch		
Title: Miata ECU daughterboard — Power		
Size: A3	Date: 2026-10-31	**Rev: 0.5b**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/7

Protected low-side drivers (NCV8406A): injectors, IACV, VVT, boost, alternator

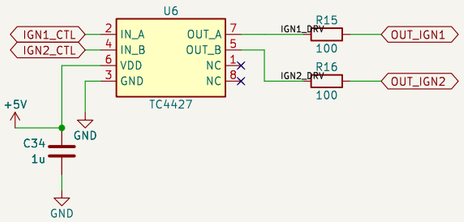


3.3 V -> 5 V level shifter for the low-side inputs (DIR low: B -> A)

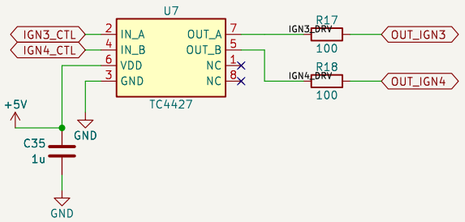


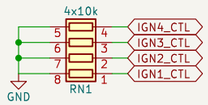
-OE pulled up: all eight drivers off until the firmware drives DRV_EN_N low

Coil-on-plug logic outputs (TC4427, 5 V push-pull through 100 R)

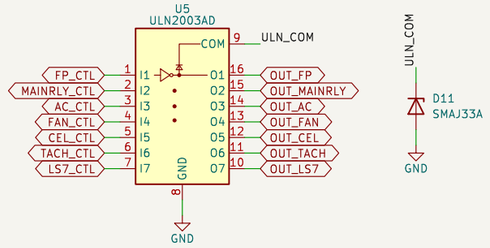


Ignition inputs pulled low (coils stay off while the MCU boots)





Relay, lamp and tach outputs (ULN2003; flyback clamp: SMAJ33A TVS on COM)



COM is not tied to +12V: a battery-fed relay coil cannot back-feed the ECU with the key off

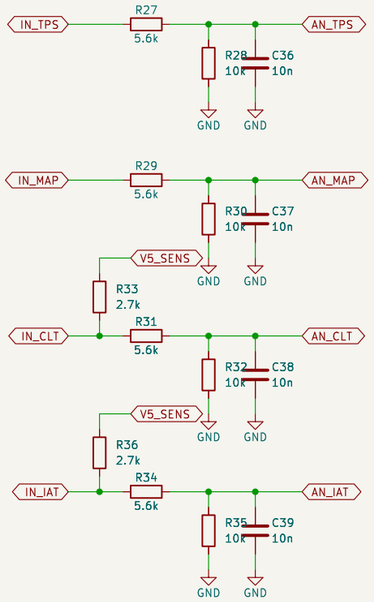
Generated from scripts/design.py; netlist identical to the routed board

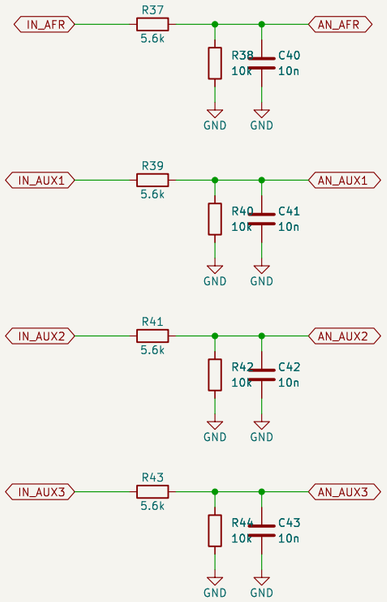
Sheet: /Outputs/
File: outputs.kicad_sch

Title: Miata ECU daughterboard — Outputs

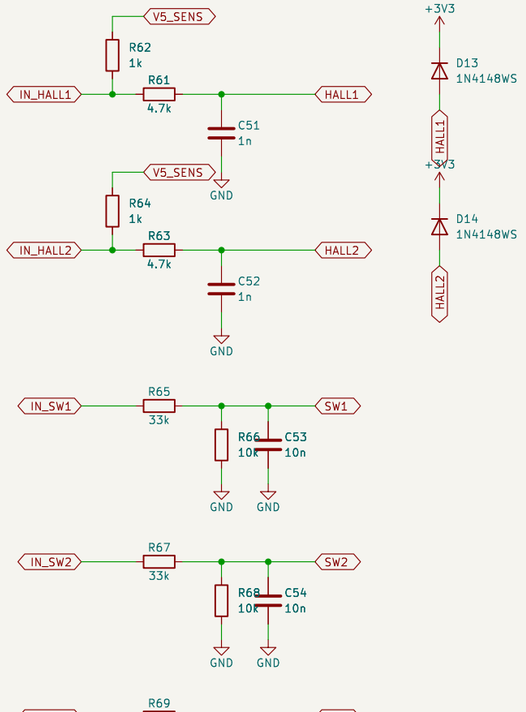
Size: A3	Date: 2026-10-31	**Rev: 0.5b**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

Analog sensor inputs: 0–5 V -> 0–3.2 V for the ADC (5.6k / 10k, 10 nF)

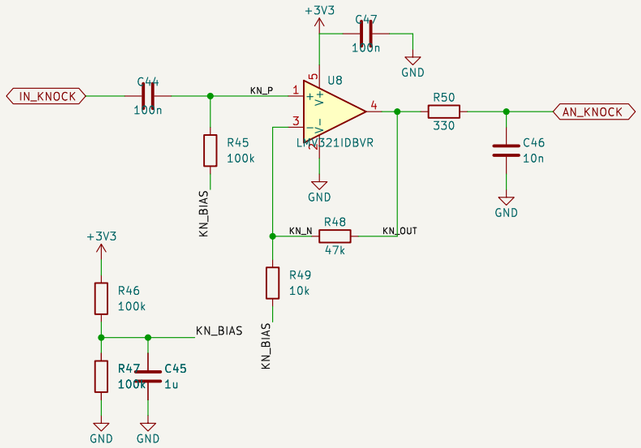




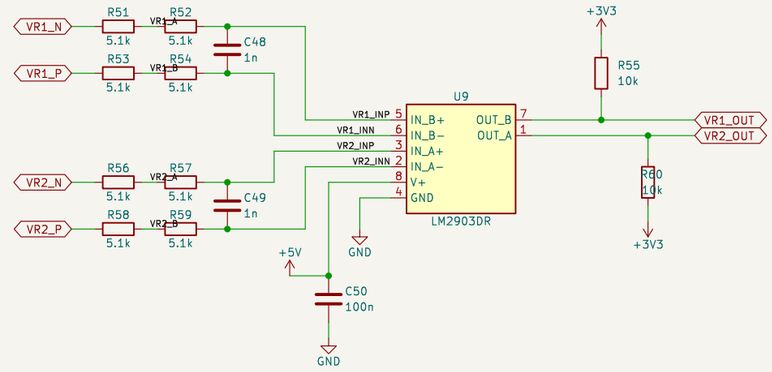
Hall (open collector, 1k pull-up) and 12 V switch inputs



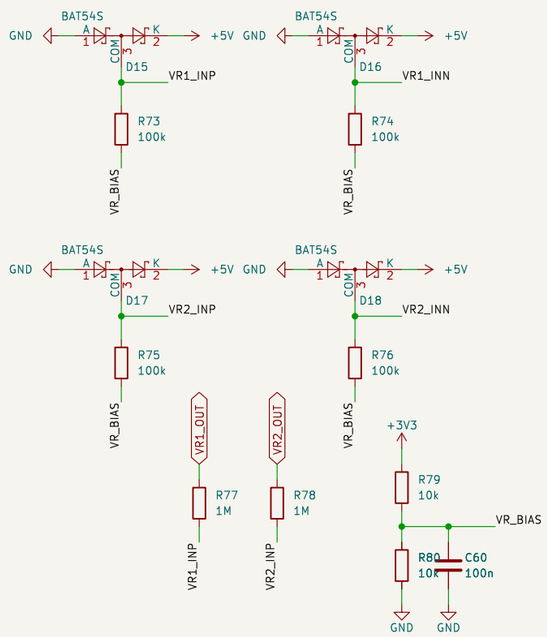
Knock: AC-coupled, biased to 1.65 V, gain 5.7



Crank / cam VR inputs (spare): LM2903 comparator, 1M hysteresis, BAT54S clamps, biased to 1.65 V



VR input clamps, bias and hysteresis



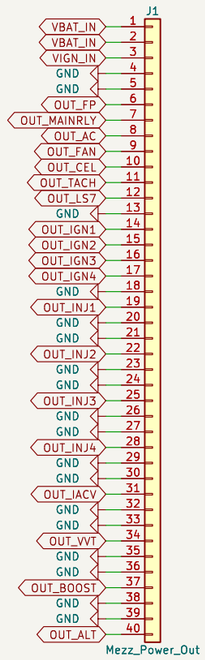
Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Inputs/
File: inputs.kicad_sch

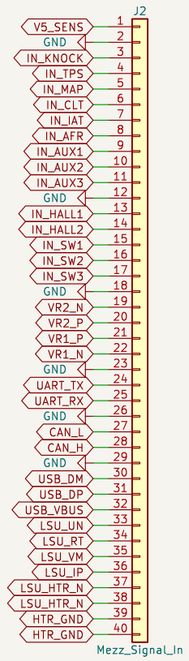
Title: Miata ECU daughterboard — Inputs

Size: A3	Date: 2026-10-31	**Rev: 0.5b**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/7

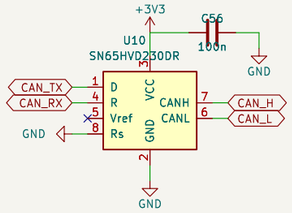
J1: power in and outputs (top edge)



J2: sensor inputs and comms (bottom edge)



CAN transceiver (termination on the motherboard)



Mounting holes



Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Connectors and CAN/
File: connectors.kicad_sch

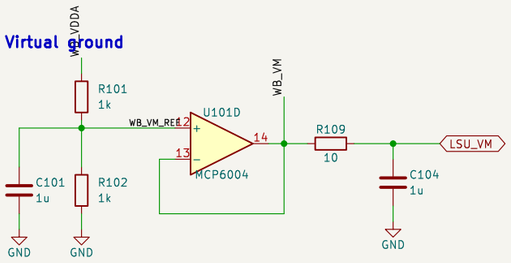
Title: Miata ECU daughterboard — Connectors and CAN

Size: A3	Date: 2026-10-31	**Rev: 0.5b**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/7

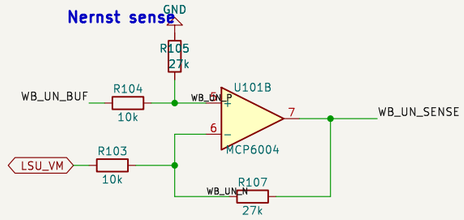
Wideband O2 controller: FOME / rusEFI open-source wideband module (MIT), rev C circuit, laid out on this board

Refs are the module refs + 100 (module R31 = R131). Talks to the main MCU over the shared CAN bus.

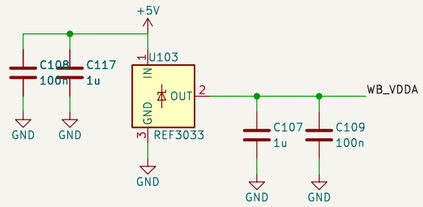
Virtual ground



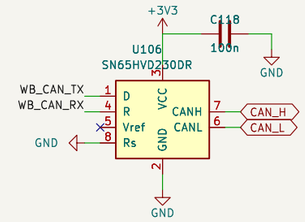
Nernst sense



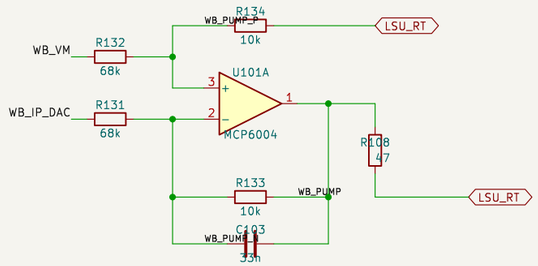
ADC / analog reference



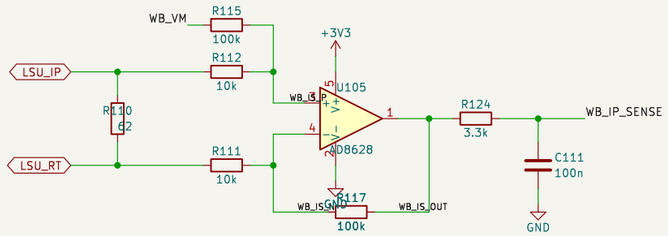
CAN (same bus as the main MCU; rusEFI reads lambda over CAN)



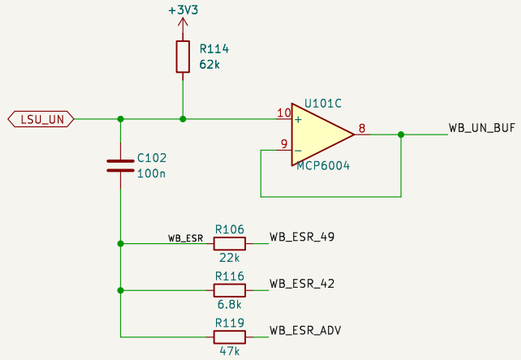
Pump current driver (Ip DAC -> LSU pump cell)



Pump current sense

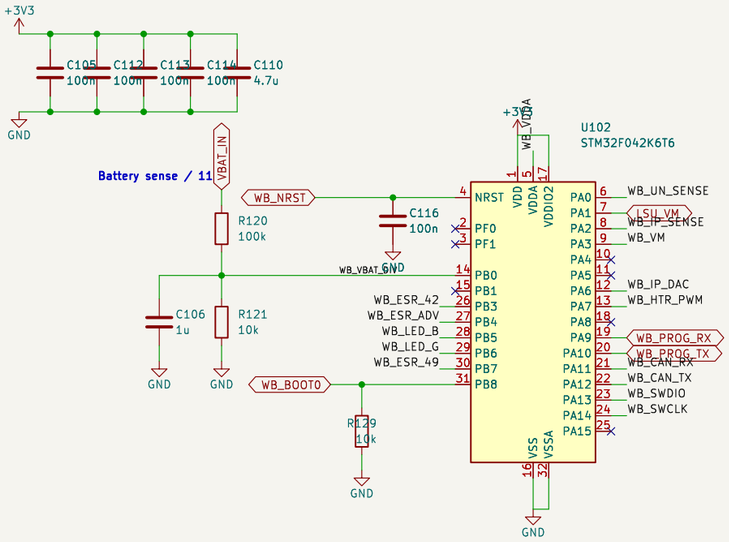


Nernst buffer, bias current and cell resistance (ESR) drive

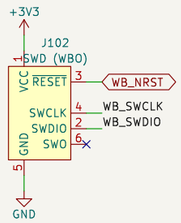


ESR drive: R106 22k / R116 6.8k / R119 47k, through C102 into the Nernst cell

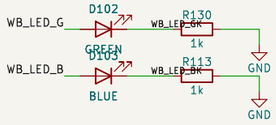
Controller MCU (STM32F042, FOME wideband firmware; first flash over SWD, then over CAN)



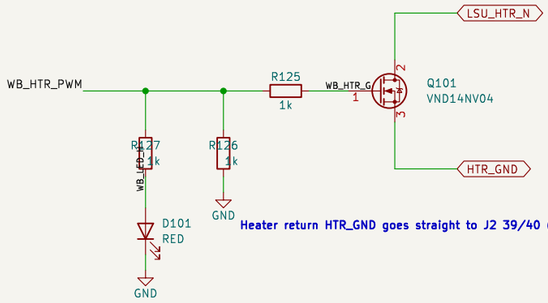
SWD (Tag-Connect TC2030-NL pads, bottom side)



Status LEDs

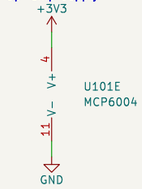


LSU heater low side (protected OMNIFET)



Heater return HTR_GND goes straight to J2 39/40 (motherboard GND), not through this board's plane

Op-amp supply



Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Wideband controller/		
File: wideband.kicad_sch		
Title: Miata ECU daughterboard — Wideband controller		
Size: A3	Date: 2026-10-31	**Rev: 0.5b**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/7